A-A-55017
May 21, 1992

COMMERCIAL ITEM DESCRIPTION

MANDRELS, MACHINE, EXPANDING

The General Services Administration has authorized the use of this commercial item description as a replacement for A-A-51129A, type II.

1. ABSTRACT

1.1 Abstract. This description covers expandable mandrels which are used for supporting workplaces between centers for machining operations.

2. CLASSIFICATION

2.1 Classification. The mandrels shall be of the sizes listed in table 1. The size furnished shall be as specified (see table I and 7.1).

3. SALIENT CHARACTERISTICS

3.1 Design. The mandrels shall be a cylindrically shaped tool consisting of a solid arbor having a specific taper over the major portion of its length and one or more expanding sleeves with an inside taper which fits the tapered portion of the arbor. The arbor shall be centered on each end with a milled flat on one end for a holding device. When the sleeve is moved along the length of the arbor, it shall contract or expand as determined by the location on the arbor. The plane of the outside surface of the sleeve shall be parallel with the horizontal center line of the arbor. The mandrels shall be similar to figure 1.

3.2 Tolerances and finish. The arbor centers shall be machined to a standard angle of 60 degrees ±30 minutes. The surface roughness average (Ra) of the arbors shall be not greater than 20 microinches using a 0.030 inch cutoff when measured in accordance with ANSI B46.1. The maximum allowable runout shall be not greater than 0.001 inch total indicator reading at the grind diameter. The sleeves shall be finished internally to fit the arbor and shall provide uniform expansion to any size in the range of diameters listed in table I.

Beneficial comments recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Defense General Supply Center, ATTN: DGSC-SS, 8000 Jefferson Davis Highway, Richmond, VA 23297-5100.

AMSC N/A
FSC 3460

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TABLE I. Sizes and dimensions.

Size	Sleeve expansion range 1/ (inches)	Arbor Length 2/ (inches)	Sleeve Length 2/ (inches)
1	3/8 to 1/2	4-1/2	1-1/4
2	1/2 to 9/16	5	2-1/2
3	1/2 to 11/16	5-3/4	1-1/2
4	9/16 to 21/32	6	2-3/4
5	21/32 to 3/4	6	2-3/4
6	11/16 to 1	9-1/4	2-1/2
7	3/4 to 7/8	7	3-1/4
8	13/16 to 1	9-1/4	2-1/2
9	7/8 to 1	7	3-1/4
10	1 to 1-3/8	9	4
11	1 to 1-9/16	12-1/4	3-1/4
12	1-1/4 to 1-5/8	9	4
13	1-1/2 to 2	11-1/2	5
14	2 to 2-3/4	14-1/2	5-1/2
15	2-3/4 to 3-3/4	17	6
16	3-3/4 to 5	20	7
17	5 to 6-1/2	24	8

1/ Where a range is shown, the dimensions shall be not greater than the lowest figure to not less than the highest figure.

2/ Tolerance of arbor length shall be ±1/2 inch and sleeve length ±l/4 inch.

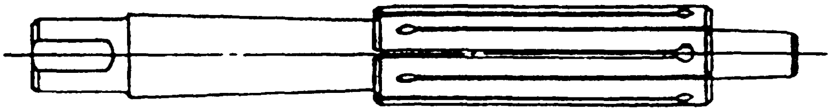


FIGURE 1. Mandrel, expanding type.

3.3 Hardness. The arbors shall have a hardness of 51 to 62 and the sleeves shall have a hardness of 38 to 47 on the Rockwell "C" scale when measured in accordance with ASTM E 18.

3.4 Identification marking. Each mandrel furnished shall be permanently marked with the sleeve size or manufacturers part number and the manufacturer's name or trademark of such known character that the source of the manufacturer may be identified.

3.4 General requirements.

3.4.1 Environmental protection. During the manufacture of the mandrels, materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect on the date of the contract shall not be used.

3.4.2 Reference documents. The documents referenced in this document shall be the issues in effect on the date of issuance of the invitation for bid or request for proposal.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent possible.

5. QUALITY ASSURANCE PROVISIONS

5.1. Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description, and that the product conforms to the producer’s own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

6. PRESERVATION, PACKING, LABELING, AND MARKING

6.1 Preservation, packing, labeling, and marking. Preservation, packing, labeling, and marking shall be as specified in the contract or order.

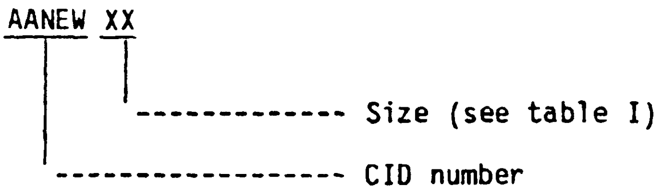
7. NOTES

7.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

a. Title, number, and date of this commercial item description.

b. Size and quantity required.

7.2 Part identification number (PIN). The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.



7.3 Source for documents referenced in this CID.

ANSI B46.1 is available from the American National Standards Institute, Attn: Sales Dept., 1430 Broadway, New York, NY 10018-3363.

ASTM E 18 is available from American Society for testing and materials, 1916 Race Street, Philadelphia, PA 19103-1187.

MILITARY INTEREST:

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Project 3460-0168

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** A-A-55017	**2. DOCUMENT DATE** *(YYMMDD)* 21 May 92

3. DOCUMENT TITLE
MANDRELS, MACHINE, EXPANDING

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

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ATTN: DIPEC-SSG

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